

Education & Economics in Kyrgyzstan

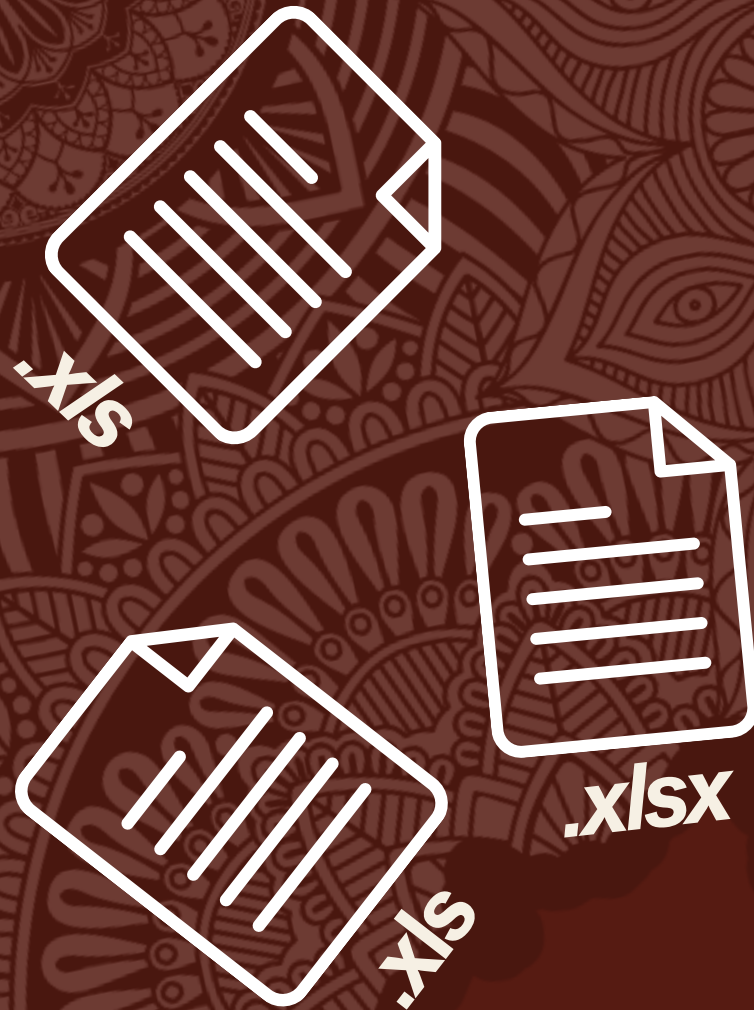


*Exploring the Relationship Between
Educational Development &
Economic Progress*

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A. Turduev*

Dataset

- Captures both education system development and economic performance
- Covers a recent 14-year period (2011–2024)



- Allows analysis of long-term trends, not short-term fluctuations
- Combines structural indicators (institutions, teachers) and financial indicators (expenditure, GDP)
- Enables investigation of education–economy relationship in Kyrgyzstan

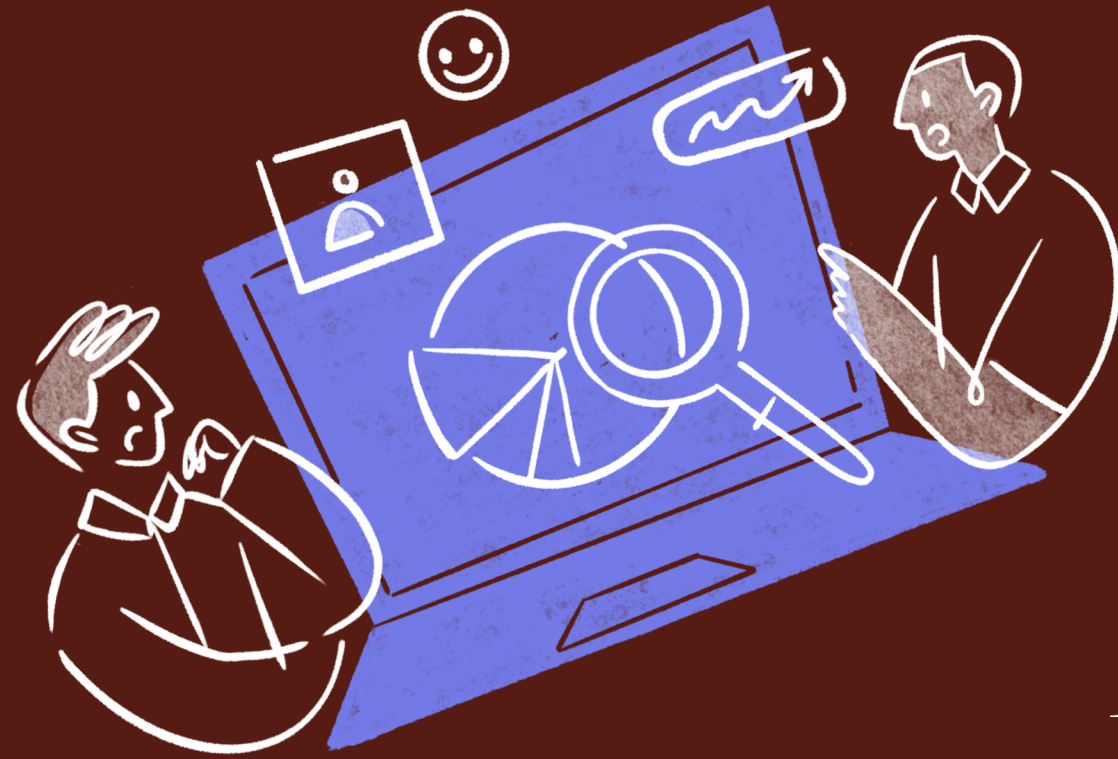
data

Objective

- Analyze the relationship between education and economic development
- Identify trends and patterns from 2011 to 2024



- Examine how education indicators relate to GDP, employment, and salaries
- Provide data-driven insights into education's role in the economy



Data Wrangling

Problems in Raw Data



Different files had
headers in different rows

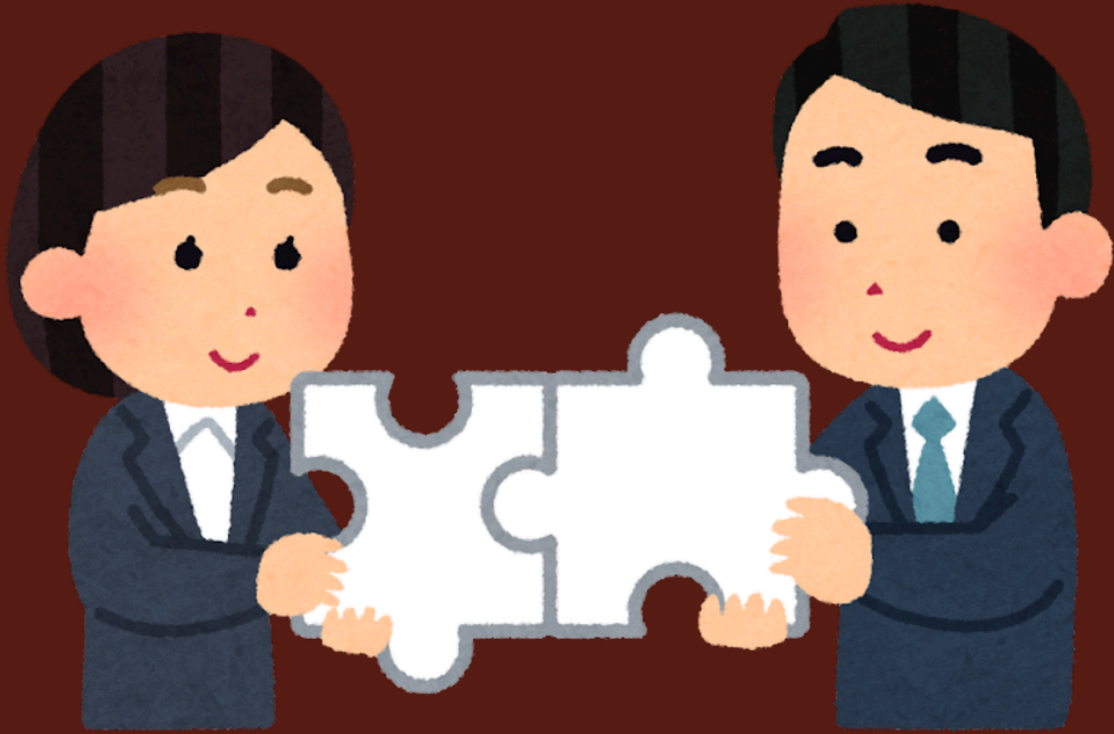


Duplicate columns existed
in multiple languages



Missing values in
several datasets

Cleaning Strategy



To ensure consistency, we implemented a reusable cleaning function applied to all datasets
It standardizes structure, converts data types, and filters the time period, making all datasets comparable

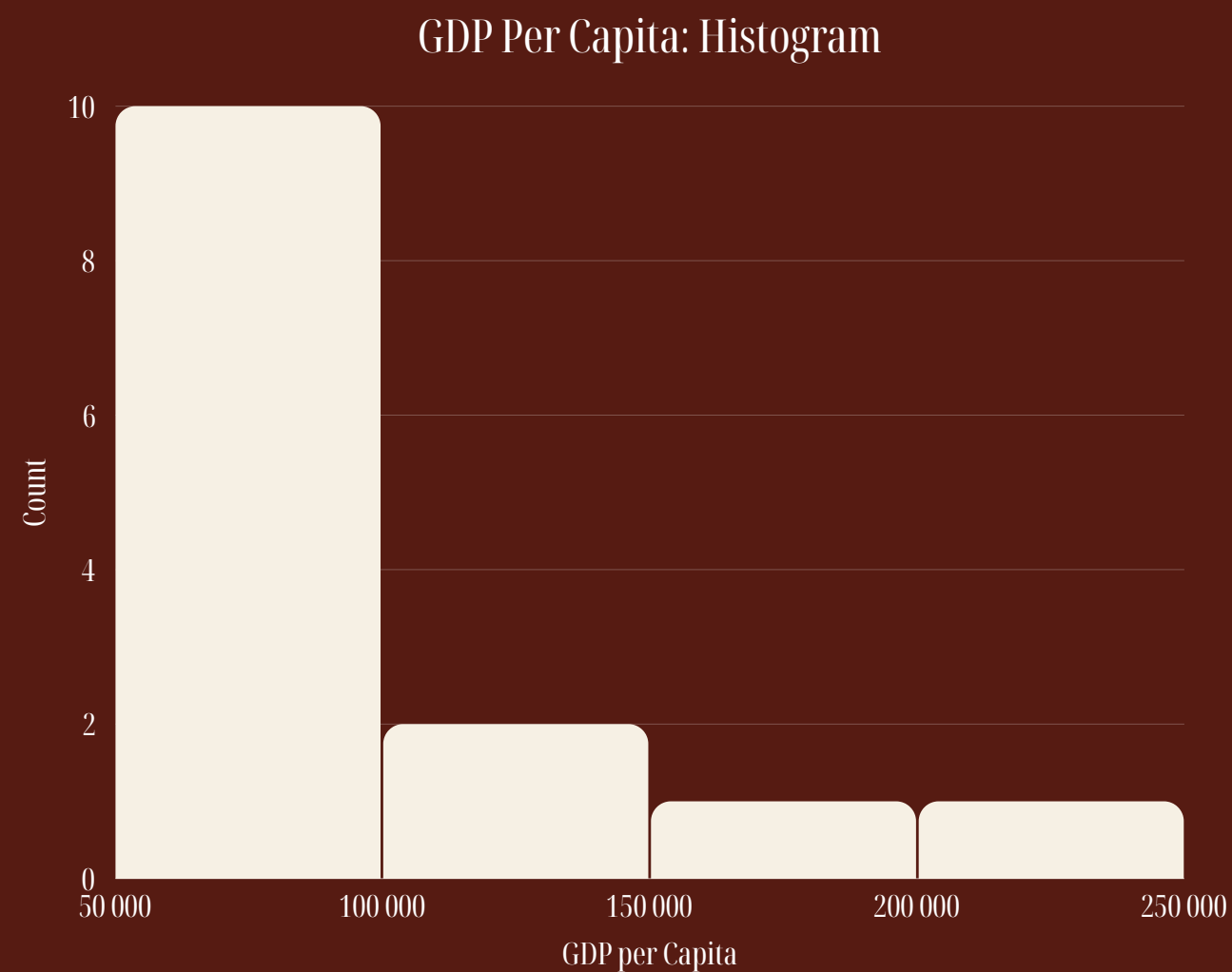
```
clean_data = dataset.copy()  
clean_data = clean_data.dropna()  
clean_data = clean_data.set_index('Items').T
```

```
clean_data.index = pd.to_datetime(clean_data.index, errors='coerce').year  
clean_data = clean_data.apply(pd.to_numeric, errors='coerce')
```

```
clean_data = clean_data[clean_data.index >= 2011]
```

Exploratory Data Analysis

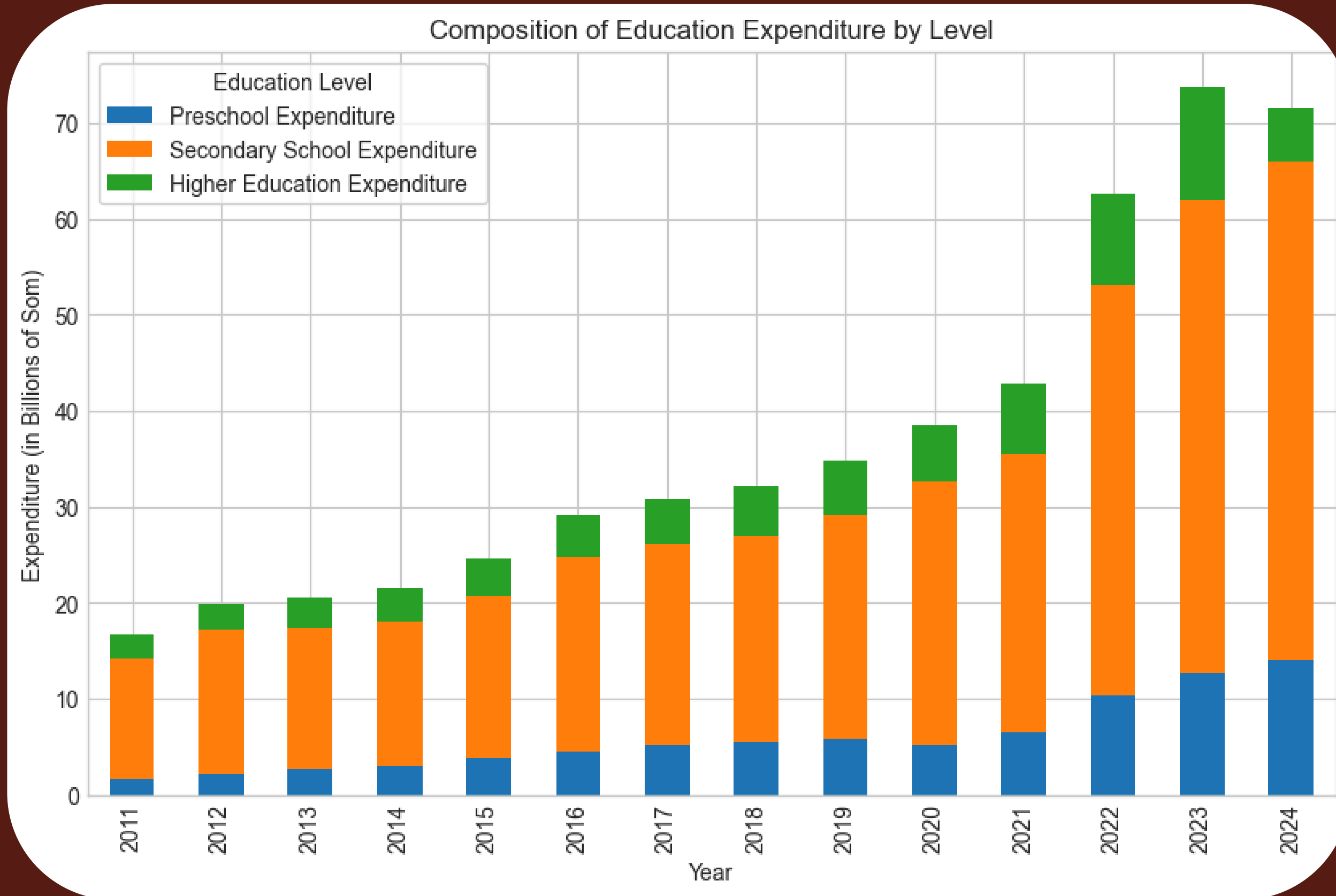
For example,



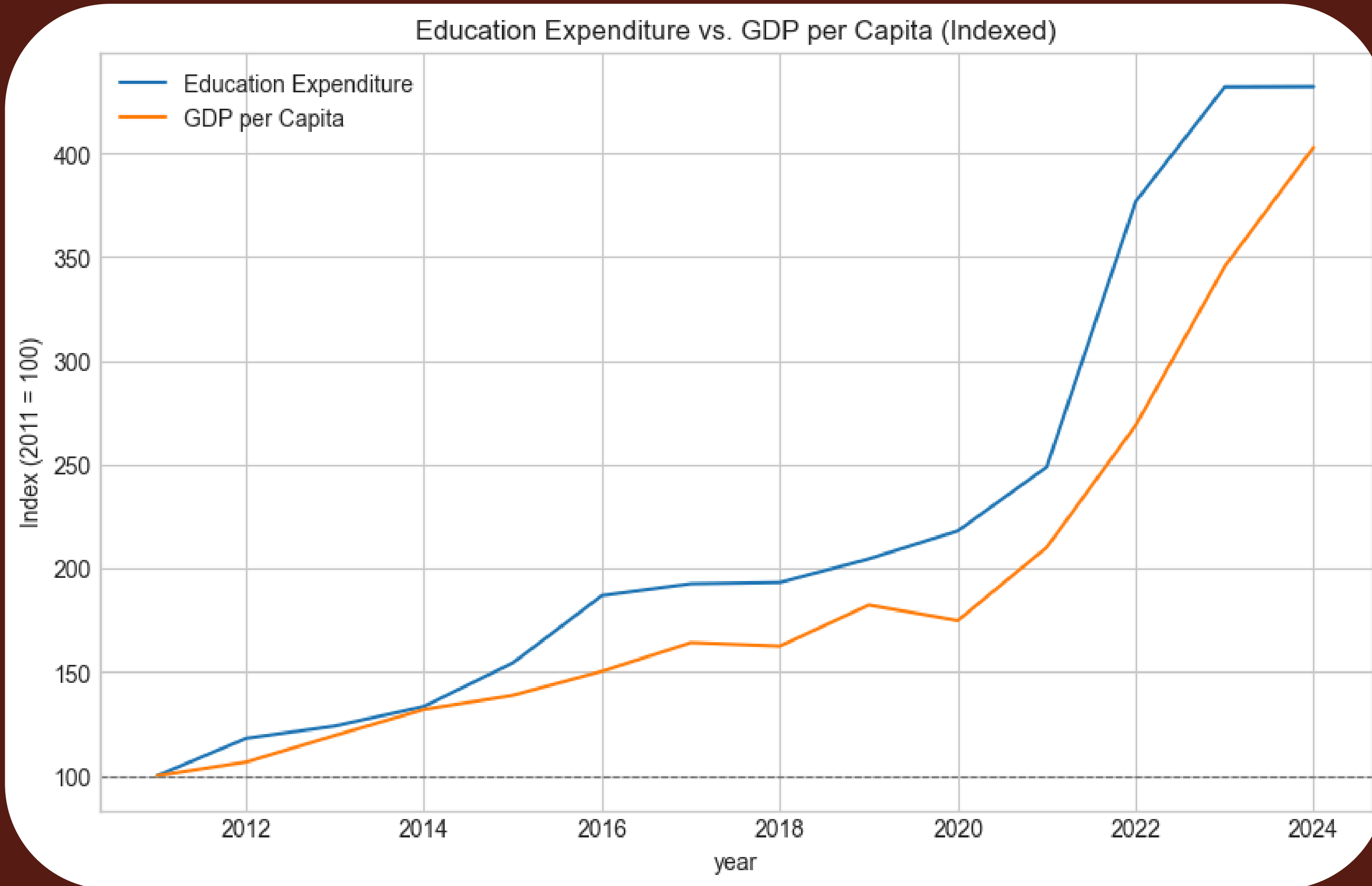
General findings:

- *Right-skewed distributions,*
- *Few outliers (GDP, salary, graduates)*

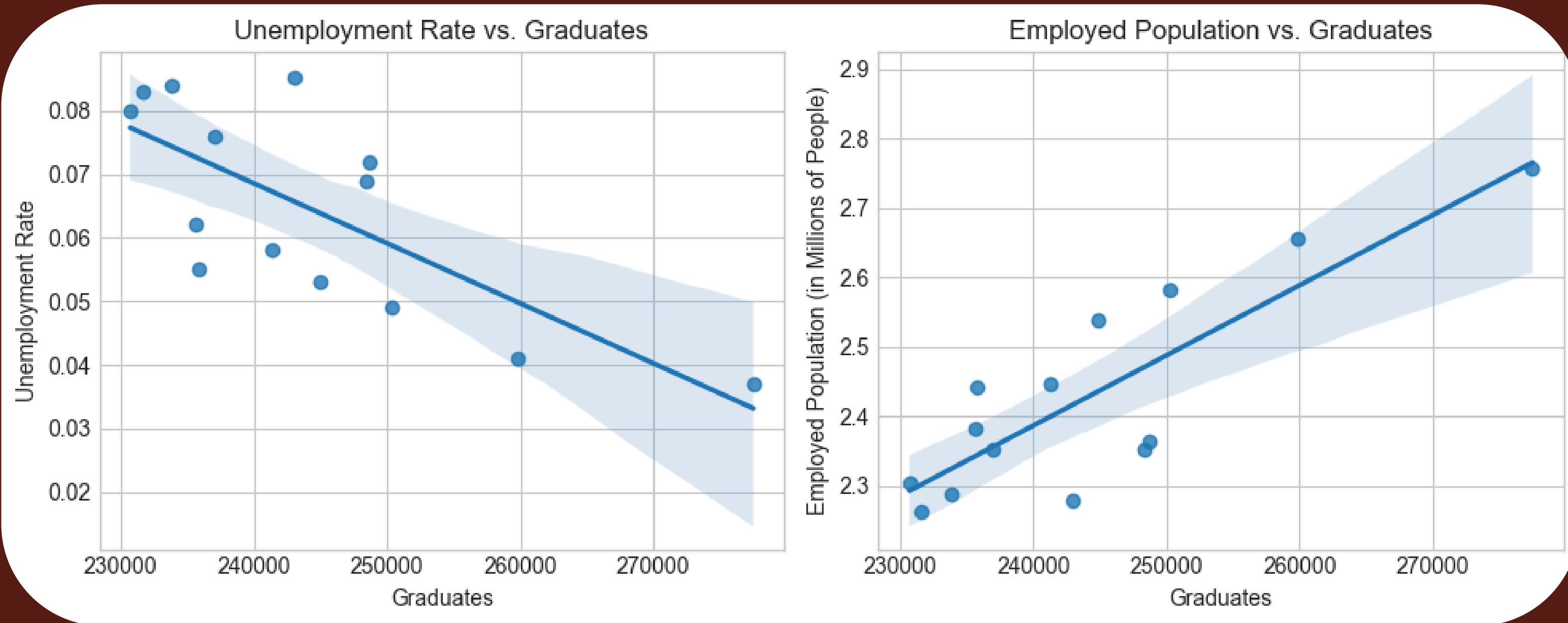
Visualizations



Visualizations

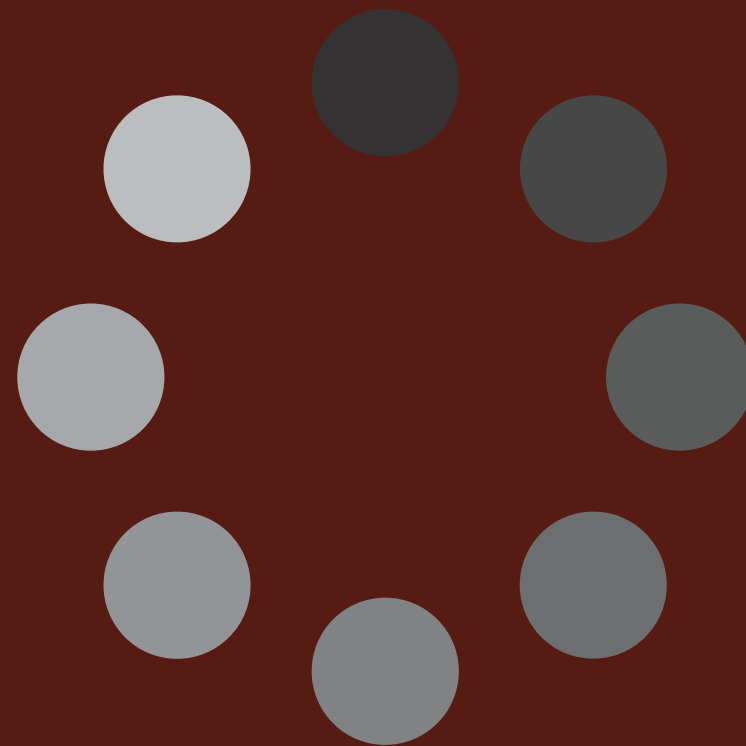


Visualizations



Dashboard

PLEASE
STANDBY



Conclusion

WE FOUND OUT:

- *Education & economics are heavily correlated;*
- *It is a two-way street.*

LIMITATIONS:

- *side-effects,*
- *absence of qualitative evaluation of educational system*